Quantifying Redundancies in Embeddings

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Abstract

Modern text embedding models often operate in high-dimensional spaces under the assumption that higher dimensionality yields better semantic representations. However, recent research demonstrates that high-dimensional text embeddings which have a large number of dimensions, often reside at a much lower dimensional manifold, exhibiting significant redundancy. In this project, we empirically study the intrinsic dimension (ID) of text embeddings derived from arXiv Computer Science abstract texts. Using k-nearest neighbors (kNN) based Local Intrinsic Dimension (LID) estimation, we quantify redundancy across embedding dimensions (ED = 50-300) and dataset sizes (5K, 20K, 50K). Motivated by low ID estimates, we evaluate PCA-based compression from 300D to 15D embeddings and benchmark retrieval performance using nDCG@10 and latency over one million queries. Our results show that the substantial speedups(~6x) are achievable with moderate retrieval quality degradation, highlighting intrinsic dimension as a principled guide for embedding compression.
Keywords: Intrinsic Dimension(ID), Extrinsic Dimension(ED), nDCG@10

1 Introduction

Vector embeddings are a core component of modern Natural Language Processing (NLP) systems, enabling tasks such as semantic search, clustering, and large-scale information retrieval. In practice, these embeddings are stored in vector databases and queried repeatedly, making both memory usage and query latency critical performance considerations.

A common assumption in representation learning is that higher-dimensional embeddings provide richer semantic representations. However, increasing dimensionality also leads to higher storage costs and slower nearest-neighbor search. Recent work has shown that embedding vectors often lie on low-dimensional manifolds embedded within high-dimensional spaces, indicating that much of the ambient embedding space may be redundant. This motivates the study of Intrinsic Dimensionality (ID), which captures the minimum number of dimensions required to describe the local geometry of the data, in contrast to the Extrinsic Dimensionality (ED) of the full embedding space.

In this work, we analyze the intrinsic dimensionality of pretrained GloVe embeddings applied to arXiv Computer Science abstracts. Using a k-nearest neighbor–based Local Intrinsic Dimension (LID) estimator, we measure ID across multiple embedding dimensions and dataset sizes. Motivated by low ID estimates, we evaluate PCA-based compression of embeddings and assess the resulting trade-offs between retrieval quality and query efficiency using nDCG@10.

1.1 Motivation

Vector embeddings are the backbone of modern Natural Language Processing (NLP) systems, powering tasks such as semantic search, clustering, recommendation and retrieval-augmented generation. These embeddings are typically stored in vector databases and queried at scale, where both latency and memory footprint are critical.

A common assumption is that higher-dimensional embeddings provide richer representations. However, this comes at a cost:

- Increased memory usage
- Slower nearest-neighbor search
- Higher computational overhead

Therefore, optimizations during NLP model training can have large effects on operational cost and efficiency, one such optimization is reducing the number of dimensions needed for model efficiency. Recent theoretical and empirical studies suggest that embedding vectors often lie on low-dimensional manifolds within a high-dimensional ambient space (Kataiwa et al.). This motivates the study of Intrinsic Dimension (ID), which is defined by Aghajanyan et al. as the free parameters that are required to closely approximate the optimization problem that is solved while fine-tuning for each end task. This is relative to the Extrinsic Dimension (ED) which is defined as the whole embedding space.

1.2 Research Questions

1) *How redundant are high-dimensional text embeddings and can intrinsic dimension estimates guide effective compression without severely degrading retrieval performance?*
2) *Are lower-dimensional representations of higher-level embeddings stable and produce consistent results?*

1.3 Background and Theory

Extrinsic and Intrinsic Dimensions:

- **Extrinsic Dimension (ED):** The ambient dimension of the embedding space. Eg: 300 for GloVe-300.
- **Intrinsic dimension (ID):** The minimum number of dimensions required to locally describe the data manifold.

When ID is very small compared to ED (ID << ED), we say that the representation is highly redundant.

Local Intrinsic Dimension (LID) measures how distances grow locally around a point. For a point x_i , let:

- $r_{i,j}$: distance from x_i to its j-th nearest neighbor.
- $r_{i,k}$: distance to its k-th nearest neighbor.

The LID-MLE estimator is:

$$\widehat{\mathrm{LID}}_k(x) = \left[\frac{1}{k-1} \sum_{i=1}^{k-1} \ln \frac{d_k(x)}{d_i(x)} \right]^{-1}$$

A global ID estimate is computed as the harmonic mean of local LID values across points.

Redundancy Ratio is defined as: $Redundancy = ID/ED$. Lower values indicate higher redundancy.

1.4 Previous Work

The concept of Intrinsic Dimension (ID) originates from manifold learning and geometric data analysis, where high-dimensional observations are assumed to lie near low-dimensional structures. Early work explored ID estimation using techniques such as correlation dimension, box-counting and maximum likelihood estimators (Levina & Bickel, 2005). These approaches demonstrated that many real-world datasets including images, text etc, exhibit effective dimensionalities far smaller than their ambient space.

Local Intrinsic Dimension (LID) further refined this idea by estimating dimensionality in a local neighborhood, capturing variations in geometric complexity across the data manifold. LID has since been applied in anomaly detection, adversarial robustness and representation analysis, where geometry provides insights beyond global statistics.

Much of contemporary work surrounding IDs are used in the context of optimizing Large Language Models (LLM). LLMs are currently one of the largest-scale applications of NLP, making optimizations in training very valuable. Janapati and Ji track how methods such as supervised fine-tuning (SFT) and in-context learning (ICL) influence the total ID space of representations in the Llama and Mistral series of LLMs. Aghajanyan et al. showed that IDs could be used to reduce the number of dimensions in LLM parameter space. The work of Ruppik et al. also explores using LID to catch problems that arise with LLMs during training such as overfitting, training limits, and grokking. Among the work presented, there is a motif of a tradeoff between model performance due to pruning dimensions and training speed, which is formally proposed in Song et al. claiming that LLM models have better performance if they have lower ED.

Much of the work presented in this paper is based on Kataiwa et al. who analyzed dimensionality reduction using intrinsic dimensions in LLM's token embedding space. Their work focused particularly on the Pythia family (14M-1.4B parameters). Using the formula kNN + LID-MLE, they studied:

- Token embedding spaces
- Hidden representation across training
- Evolution and stabilization of ID over training steps.

Their results showed that ID stabilizes early in training and grows sublinearly with model size, suggesting substantial redundancy in representation spaces.

Our work differs in scope:

- We study fixed, pretrained embeddings (GloVe) rather than evolving model representations.
- We focus on retrieval efficiency, not training dynamics.
- We connect ID estimates to practical compression and search performance.

2 Methods

We carry out our experiments on token and sentence level embeddings. Both of these experiments follow the same robust method from the source paper Kataiwa et al. The method involves estimating the LID and Global ID.

1. LID Estimation: We estimate the LID of any given vector **x** using the Maximum Likelihood Estimator from Levina and Bickel:

$$\widehat{\mathrm{LID}}_k(x) = \left[\frac{1}{k-1} \sum_{i=1}^{k-1} \ln \frac{d_k(x)}{d_i(x)} \right]^{-1}$$

2. Global ID Estimation: the Global ID of an entire embedding will be computed as the harmonic mean of all the individual LID values.

$$\widehat{\mathrm{ID}} = \left[\frac{1}{n} \sum_{i=1}^{n} \widehat{\mathrm{LID}}_i^{-1} \right]^{-1}$$

   We use the FAISS library for efficient k-nearest neighbor search as recommended by the paper Kataiwa et al.

3. Quantifying redundancy: redundancy is quantified by the ratio:

$$\mathrm{Redundancy} = \frac{\mathrm{ED} - \mathrm{ID}}{\mathrm{ED}}$$

2.1 Dataset and Preprocessing

The dataset used in this study is derived from the arXiv metadata corpus, a large, publicly available collection of scientific paper metadata spanning multiple research domains. From this

corpus, we restrict our analysis to Computer Science categories ensuring a relatively homogeneous semantic domain while maintaining substantial lexical and topical diversity.

Each data point corresponds to the abstract text of a paper, which provides a concise but information-dense summary of the research content. Abstracts are particularly suitable for embedding-based analysis, as they contain sufficient semantic structure while remaining computationally manageable.

We process the raw arXiv metadata dump (JSONL format) and apply the following filters:

- Retain only the Computer Science category.
- Discard records with missing or empty abstracts.
- Store each abstract as a single line of plain text.

2.2 Dataset Sampling

To study the effect of dataset size on intrinsic dimension estimation, we create three independent datasets by uniform random sampling of sizes 5000, 20000 and 50000. These sample sizes are chosen to balance statistical robustness with computational feasibility from the view of exact kNN searches required by intrinsic dimension estimation.

2.3 Embedding Models

We use pretrained GloVe embeddings which provide fixed vector representations for individual tokens based on global co-occurrence statistics. The embedding models evaluated are GloVe-50, GloVe-100, GloVe-200 and GloVe-300.

Token level embeddings:

- Each unique token appearing in the dataset is mapped to its corresponding GloVe vector.
- Tokens not present in GloVe vocabulary are discarded.
- The resulting matrix forms a token level point cloud.

Abstract level embeddings:

- Each abstract is represented by the mean of its in-vocabulary token embeddings.
- Abstracts with no valid tokens are discarded.
- This yields a second point cloud with one point per abstract.

2.4 Internal Dimension Estimation

To quantify redundancy in high-dimensional text embeddings, we estimate the intrinsic dimension (ID) of the embedding space using a k-nearest neighbor (kNN)–based local intrinsic dimension (LID) estimator. This section describes how embeddings are represented

geometrically, how neighborhood distances are computed, and how stability across neighborhood sizes is ensured.

For the purpose of intrinsic dimension estimation, embedding representations are interpreted geometrically as point clouds in a Euclidean space. Specifically, each embedding matrix is treated as a collection of points in a high-dimensional ambient space. Rows in the data represent token level embeddings in token-level analysis and abstracts embeddings in abstract-level embeddings. The number of columns equals the extrinsic dimensions(ED) which varies from 50, 100, 200 to 300. This geometric interpretation allows us to analyze the local neighborhood structure of the embeddings, which is essential for intrinsic dimension estimation. Importantly, repeated occurrences of the same token are excluded at the token level, ensuring that the point cloud reflects unique geometric locations rather than frequency effects.

Local intrinsic dimension estimation relies on distances between each point and its nearest neighbors. To compute these distances efficiently and accurately we have performed exact k-nearest neighbor searches using the FAISS library.

- We use FAISS IndexFlatL2, which performs brute-force exact nearest neighbor search under the Euclidean (L2) distance metric.
- For each point x_i, we retrieve its k+1 nearest neighbors and discard the first neighbor which corresponds to the point itself (self-match).
- The remaining k distances $\{r_{i,1},\ldots,r_{i,k}\}$ form the local neighborhood used for LID estimation.

To assess robustness, we compute kNN distances for multiple neighborhood sizes: $K \in \{10,20,50,100,200\}$. This range spans small to moderately large neighborhoods, capturing both fine-grained local geometry and more global structure.

The choice of neighborhood size k plays a crucial role in LID estimation. Very small values of kkk may lead to high variance due to noise, while very large values may blur local geometric structure by incorporating non-local neighbors.

To ensure that intrinsic dimension estimates are not artifacts of a particular neighborhood size, we perform a stability analysis across multiple values of k:

- For each dataset and embedding dimension, ID is estimated independently for each k.
- We analyze how global ID varies as a function of k.
- Stable plateaus in the ID vs. k curves indicate reliable intrinsic dimension estimates.

By representing embeddings as point clouds, computing exact kNN distances, and validating estimates across multiple neighborhood sizes, we obtain reliable and interpretable intrinsic dimension estimates. These estimates form the foundation for subsequent analyses of redundancy and compression-based retrieval efficiency.

3 Results

We begin by analyzing how intrinsic dimension (ID) varies with extrinsic dimension (ED) across different dataset sizes. For each dataset (5k, 20k, and 50k abstracts), we compute the global intrinsic dimension using the LID-MLE estimator and report the ratio ID/ED.

We found that the redundancy increases as the ED space increases as we increase the size of the embeddings. As shown in Figures 1, 2, and 3 this correlation is repeated across all of our token sample sizes (5k/20k/50k) and at 300 dimensions, the observed redundancy ratio is about 0.05 to 0.07, meaning the effective number of dimensions to achieve convergence is 15 to 20. The y-axis represents the Redundancy Ratio(ID/ED) and the x-axis represents the size of embeddings used for experimentation(50, 100, 200 and 300).

Some observations are:

- Across all dataset sizes, the ID/ED ratio decreases as ED increases, indicating increasing redundancy in higher-dimensional embeddings.
- For ED = 300, the ID/ED ratio consistently falls in the range 0.05–0.07, suggesting an effective dimensionality of approximately 15–20.
- The similarity of curves across dataset sizes indicates that these trends are not artifacts of sample size, but reflect underlying geometric structure of the embedding space.

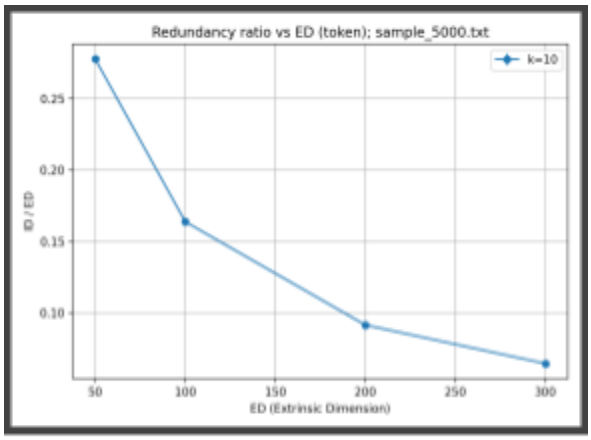


Fig 1. Redundancy Ratio with ED for sample size = 5000.

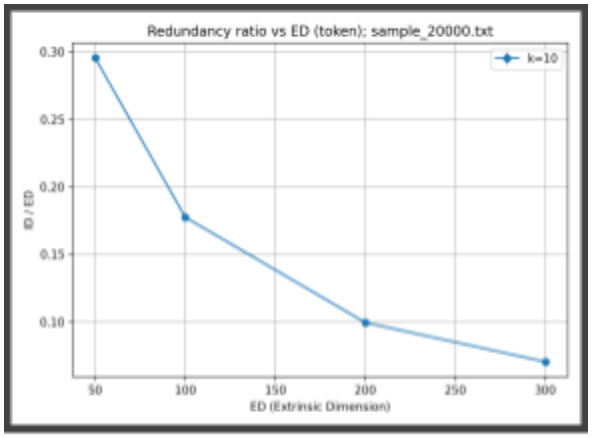


Fig 2. Redundancy Ratio with ED for sample size = 20000

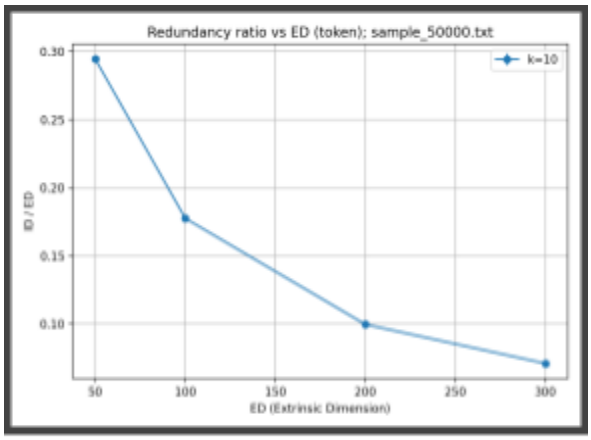


Fig 3. Redundancy Ratio with ED for sample size = 50000.

Because intrinsic dimension estimation depends on the neighborhood size k, we assess the robustness of ID estimates by varying k over a wide range. The results indicate ID tends to converge on the model earlier and produce more stable results during training in models with higher ED. Figure 4 shows the stability of ID as the threshold k increases, reaching a stable number of dimensions when the faded outline disappears. The x-axis corresponds to the number of nearest neighbors k. The y-axis shows the estimated global intrinsic dimension. Separate curves represent different embedding dimensions.

Some observations are:

- For small values of k, ID estimates exhibit high variance, reflecting sensitivity to local noise.
- As k increases, ID estimates stabilize into a plateau, typically around k≈70.

- For very large k, slight downward drift is observed, consistent with the loss of locality when neighborhoods become overly global.

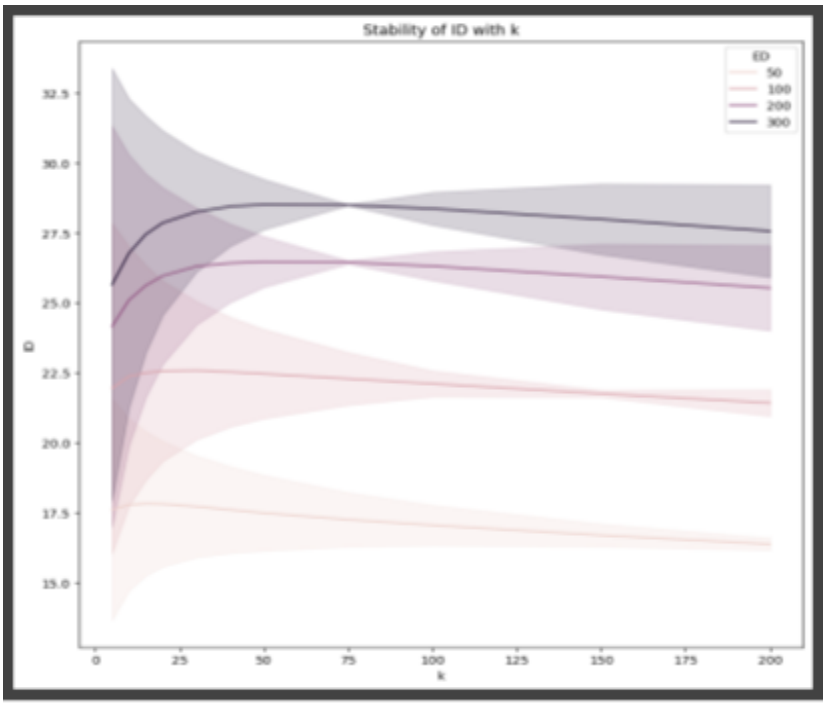


Fig 4. Stability of Internal Dimensions with k

Combining the ID/ED trends with stability analysis, we observe that:

- Intrinsic dimension grows sublinearly with extrinsic dimension.
- Increasing ED beyond a certain point yields diminishing returns in representational complexity.
- For ED = 300, the effective dimensionality is close to 15–20, suggesting substantial redundancy.

These observations motivate a natural hypothesis that compressing 300D embeddings to approximately the intrinsic dimension should preserve much of the retrieval structure while significantly improving efficiency. To test this hypothesis, we conduct a large-scale retrieval benchmark comparing uncompressed and compressed embeddings.

Setup for Compression-Based Retrieval Benchmark:

- Database: 50000 abstract embeddings (ED = 300).
- Compression: PCA from 300D to 15D.
- Queries: 1,000,000 noisy queries derived from database points.
- Vector search: exact cosine similarity using FAISS.

- Evaluation metric: nDCG@10, with relevance defined by the top-100 neighbors in the 300D space.

We evaluate how compression affects retrieval quality using nDCG@10. The benchmark to compute the nDCG@10 is the uncompressed embeddings and then we check how closely are we able to capture the same elements with compressed embeddings.

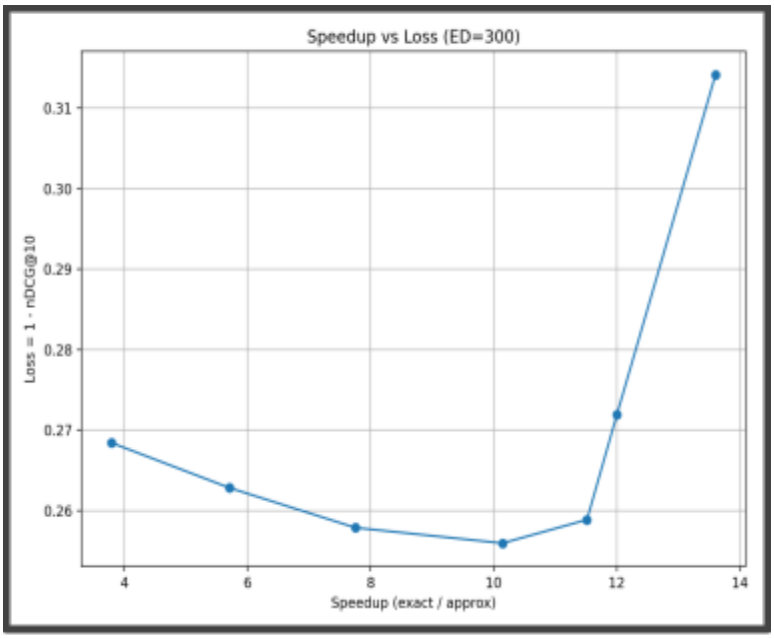


Fig 5. Vector Search Query vs Loss of information for original ED = 300.

Some observations are:

- PCA compression to 15D achieves an average nDCG@10 ≈ 0.69.
- This corresponds to a ranking quality loss of approximately 30% relative to the ideal 300D retrieval.
- As seen in Figure 5, where x-axis represents the Speedup of queries and y-axis represents how much information is lost for each speedup factor. As we increase speedup(meaning we compress more), initially the loss reduces showing that the embeddings go to the best manifold and after that compressing them further means breaking the local geometry and hence affecting the loss negatively. The plot shows that there is a sweet spot at Speedup of 10 times where we minimize the loss and also speedup our search significantly.

4 Discussion

Our results show that intrinsic dimension estimates for text embeddings are stable across dataset sizes, indicating that intrinsic dimension captures an inherent geometric property of the

embedding space rather than an artifact of sampling. Across datasets of 5k, 20k and 50k arXiv CS abstracts, embeddings with an extrinsic dimension (ED) of 300 consistently exhibited an intrinsic dimension in the range of 15–20, in agreement with prior findings by Kataiwa et al. Additionally, the observed decrease in the ID/ED ratio as ED increases was consistent across all dataset sizes, suggesting that redundancy increases with embedding dimensionality and that intrinsic dimension is largely invariant to dataset size given sufficient coverage.

We further validated the robustness of intrinsic dimension estimation through a stability analysis across neighborhood sizes k. ID estimates were noisy for small k, stabilized over a moderate range ($k \approx 70$), and showed mild drift for very large k, consistent with known properties of LID-MLE estimators. This analysis confirms that the reported ID values are not sensitive to a particular choice of neighborhood size and supports the reliability of intrinsic dimension as a summary measure of embedding geometry.

Motivated by the low intrinsic dimensionality of the embeddings, we evaluated the practical implications of redundancy through a large-scale retrieval benchmark. Compressing 300-dimensional embeddings to 15 dimensions using PCA yielded an approximately 6× end-to-end speedup in retrieval latency while preserving a substantial portion of ranking quality ($\text{nDCG@10} \approx 0.69$). These results demonstrate that intrinsic dimension provides a principled basis for dimensionality reduction in retrieval systems, enabling meaningful efficiency gains with controlled quality loss and suggest that ID-aware compression strategies can play an important role in the design of scalable vector search systems.

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